Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **Python and SQL data platforms**, **REST APIs**, validation workflows, schema-driven systems, automation, and reliable research-oriented pipelines.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reduced dashboard latency from 1.2s to 350ms through **SQL indexing and query optimization**, designing normalized schemas and row-level isolation for a multi-tenant education platform supporting 500+ users reliably.
- Achieved 99%+ production reliability across 5K+ monthly jobs through **automated data validation and reconciliation**, error handling, cloud storage, CI/CD releases, and stable observability for dependable downstream workflows.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous PDF, DOCX, and ZIP inputs while isolating failures and preserving responsive user workflows during concurrent processing reliably.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Improved long-horizon trajectory planning by 18% in simulation by building a **structured data pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and reproducible evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins and GitHub**, adding failure reports, data checks, and cross-platform validation with repeatable nightly workflows across production environments reliably and consistently.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Synchronized 2M+ records through an eight-service **resource-management platform** covering authentication, allocations, customers, billing, permissions, and notifications with Node.js, WebSockets, and Azure SQL across distributed locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing role-based access, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

Skills

Languages: Python, SQL, Java, TypeScript, JavaScript, Bash, C++
Data & APIs: REST APIs, Data Validation, Data Pipelines, Schema Design, Query Optimization, Data Modeling, ETL, JSON, WebSockets
Databases & Storage: PostgreSQL, MySQL, AWS RDS, AWS S3, Azure SQL, MongoDB, Redis, Firebase
Engineering & Automation: Git, GitHub, Docker, Jenkins, CI/CD, Command-Line Tools, Scripting, Scheduled Jobs, Workflow Automation
Quality & Support: Unit Testing, Integration Testing, PyTest, Mockito, Production Debugging, Error Handling, Logging, Observability, Documentation, Role-Based Access Control

Projects

Laboratory Metadata Platform - Design Concept

- Designed a **Python and PostgreSQL lab-metadata** concept with experiment templates, sample lineage, schema migrations, validation rules, role-based permissions, audit history, REST APIs, and reviewable change workflows for reliable records.

Research Data Access CLI - Design Concept

- Designed a **Python command-line workflow** concept for querying experiment metadata, validating inputs, exporting analysis-ready records, scheduling recurring checks, logging failures, generating runbooks, and safely retiring repetitive manual requests.

Chat2SQL | Live

- Unified **10+ database systems** in a Java and TypeScript workspace extending Chat2DB, integrating schema exploration, SQL generation, structure synchronization, and Docker-based local deployment across Windows, macOS, and Linux environments.